MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING,

30 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 81, 81.1, 345.2 AND 345.3

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

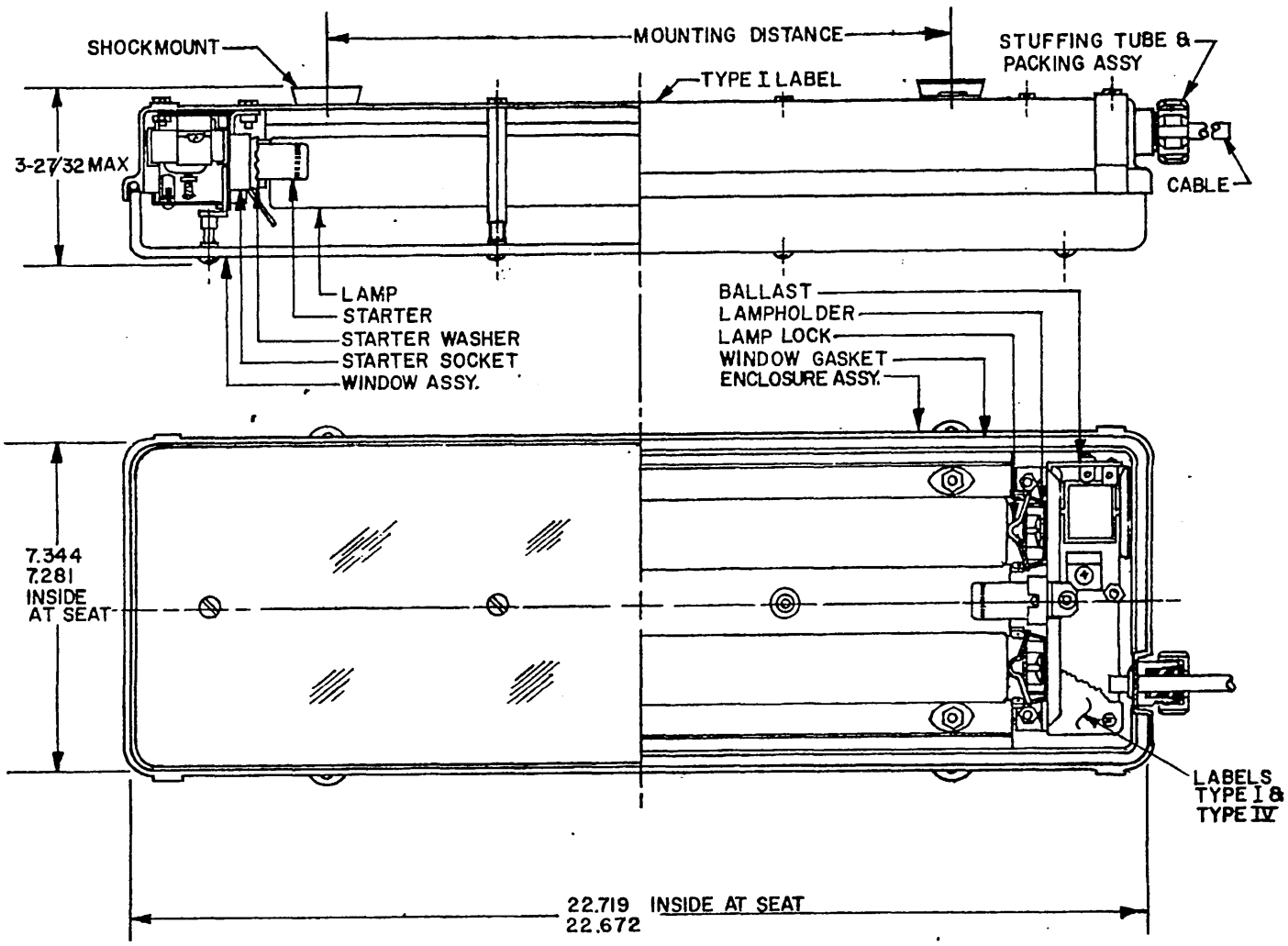


FIGURE 1. Typical view of Symbols 81 and 345.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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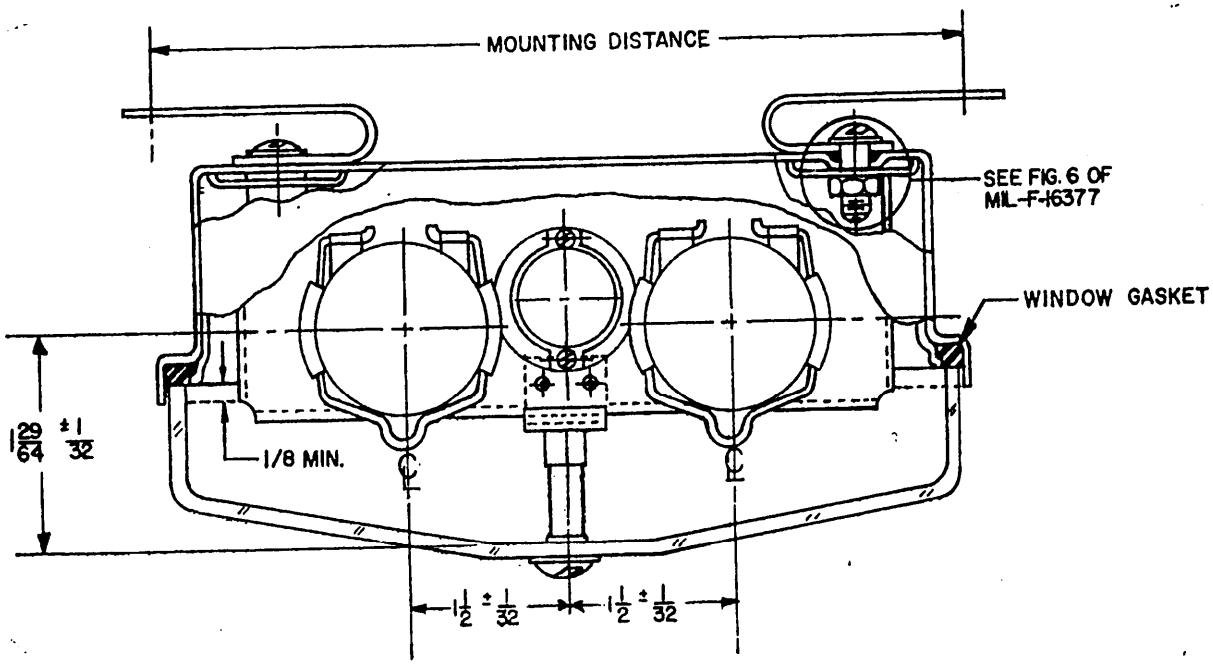


FIGURE 2. Typical section for Symbols 81 and 345.2 .

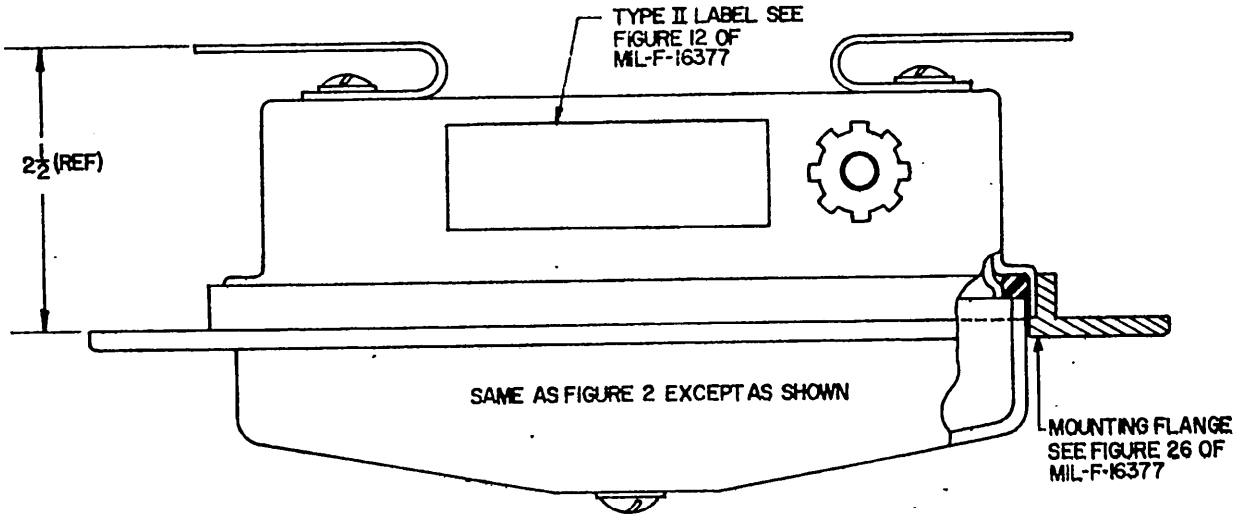
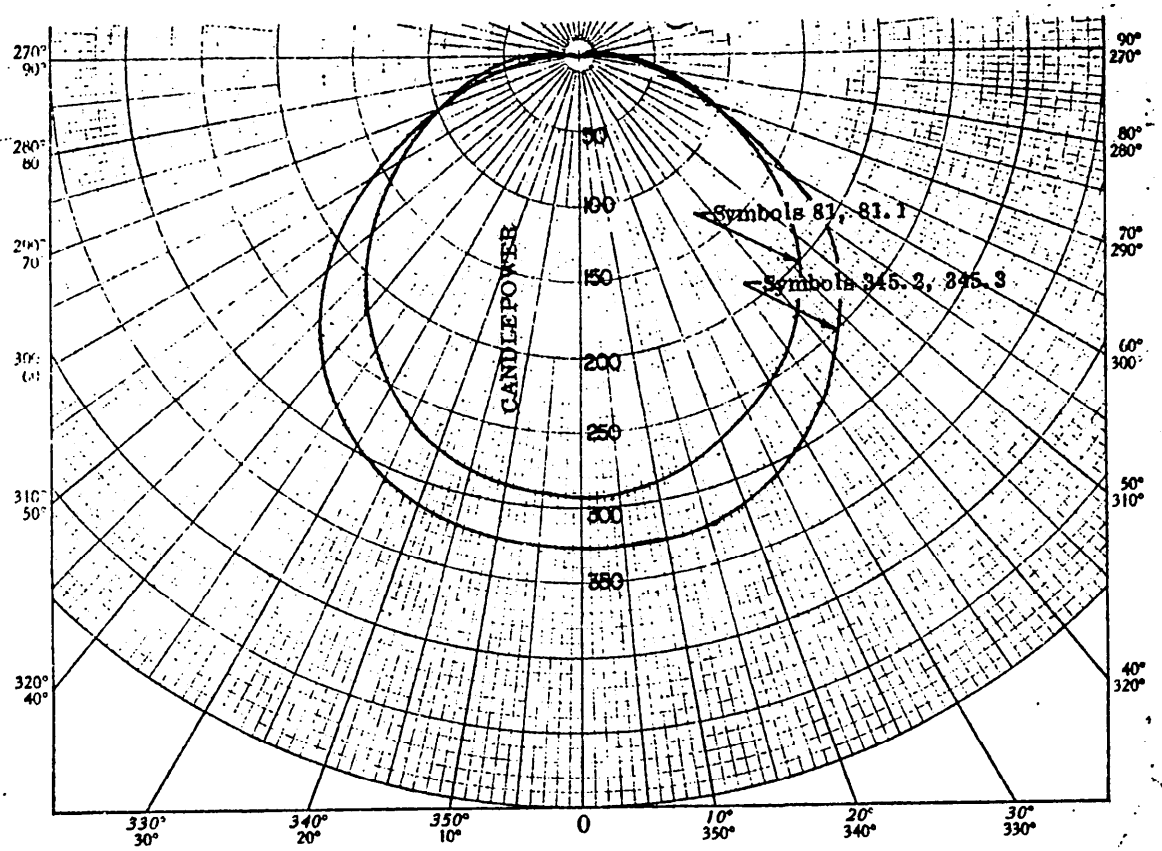


FIGURE 3. Typical section for Symbols 81.1 and 345.3.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	81, 81.1	345.2, 345.3
0	290	320
5	290	320
10	285	320
15	280	320
20	275	315
25	265	305
30	255	290
35	235	280
40	222	265
45	205	245
50	177	225
55	160	195
60	140	160

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	81, 81.1	345.2, 345.3
65	120	130
70	95	95
75	75	75
80	55	55
85	42	40
90	25	25
95	10	16
100	0	10
105	0	0
110	0	0
115	0	0
120	0	0
125	0	0

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 790 lumens per lamp.

FIGURE 4. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
81: Complete lighting fixture assembly, standard mounting, with two 15-watt cool white fluorescent lamps and white diffusing window assembly.
81.1: Complete lighting fixture assembly, flush mounting with two 15-watt cool white fluorescent lamps and white diffusing window assembly.
345.2: Complete lighting fixture assembly, standard mounting with two 15-watt cool white fluorescent lamps and clear and white prismatic window assembly.
345.3 Complete lighting fixture assembly, flush mounting, with two 15-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 30 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2 and 3 and figure 10 of MIL-F-16377.
Photometric data: see figure 4.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 23 of MIL-F-16377.
Mounting dimensions:
Symbols 81 and 345.2: see figure 25 of MIL-F-16377.
Symbols 81.1 and 345.3: see figure 26 of MIL-F-16377.
Window assembly:
Symbols 81 and 81.1: Military Part Number M16377/65-001, figure 27 of MIL-F-16377.
Symbols 345.2 and 345.3: Military Part Number M16377/65-002, figure 27 of MIL-F-16377.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet of type SJO or SJT, in accordance with J-C-580 except, AWG size 16, three conductor (white, black and red).
Weight:
Symbols 81 and 345.2: 8 lbs 11 oz. maximum.
Symbols 81.1 and 345.3: 10 lbs 1 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I.
(b) One type II for Symbols 81.1 and 345.3 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in Table I and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.
Quality conformance: Same as the first article inspection.

Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/13	6240-00-583-3668	Industry Number F15T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	MIL-L-970/11	6250-00-284-0481	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, Type III of W-S-755	6250-00-299-2884	Commercial Designation: FS2
Ballast	M16377/44-001	6250-00-569-9502	
Shockmount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-004	6210-00-834-1621	Not furnished See 3.7.7 of MIL-F-16377

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military Part number Ml6377/65-	Item name	National Stock Number (NSN)	Remarks
81	Complete fixture, Symbol 81		New
81.1	Complete fixture, Symbol 81.1		New
345.2	Complete fixture, Symbol 345.2		New
345.3	Complete fixture, Symbol 345.3		New
001	Window assembly, white diffusing		New
002	Window assembly, clear and white prismatic		New

Preparing activity:
Navy - SH
(Project 6210-N569)

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